DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot C

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-C06: Murder rate (per 100,000, 1976) [20 marks]

Brief: You are building a map story about Murder rate (per 100,000, 1976) across the United States for a state planning commission. Your comparison variable is Population (thousands, 1975). Your focus region is the Northeast region. Your map uses 6 classes and the "inferno" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$murder      # the variable you map
W <- d$pop      # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of murder: ggplot(mapdat, aes(long, lat, group = group, fill = murder)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "inferno") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 6 classes: equal interval, cut(V, 6), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of murders: count <- d$pop * 1000 * d$murder / 100000. Name the top 3 states by count and the top 3 states by murder [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the Northeast region. Report the mean of murder in Northeast and its ratio to the mean of all 50 states [1]; the highest state in Northeast with its value [0.5]; and how many Northeast states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a state planning commission [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

Code of Conduct: I certify that this work is my own. Signature: _____

LAB EXPERIMENT RECORD

Map Storytelling Challenge MS-C06: Murder rate (per 100,000, 1976)

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Map Storytelling Challenge MS-C06 - Murder rate (R)
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	22 OCTOBER 2026
Day	Thursday

Aim

To build a map story about the murder rate (per 100,000 residents, 1976) across the United States for a state planning commission: to draw a choropleth, to compare the four regions, to compare two classification methods with 6 classes, to show why counts mislead against rates, and to tell the story of the Northeast with a headline, two evidence sentences and a recommendation.

Story Brief

The story is built from the R dataset state.x77 (50 states). The mapped variable V is the murder rate (per 100,000, 1976) and the comparison variable W is the population (thousands, 1975). The focus region is the Northeast and the map uses 6 classes and the inferno colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Task	Content	Marks
1	Base map: choropleth, highest, lowest and median	4
2	The regional story: regional means and cor(V, W)	4
3	Classes change the story: equal interval against quantile	4
4	Counts versus rates	4
5	Tell the story of the Northeast	4

All code below was executed in R (ggplot2 and maps) and each block is the real console transcript: the lines starting with '>' are the commands run and the lines under them are the output R printed. The figures were produced by the same code, so every number quoted in the observations comes from the output shown.

Set-up (run first, not edited)

Program and Output

```
> library(ggplot2); library(maps)

> d <- data.frame(state = state.name, zone = state.region, state.x77)

> names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")

> d$id <- tolower(d$state)
```

```
> mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")

> mapdat <- mapdat[order(mapdat$group, mapdat$order), ]

> V <- d$murder   # the variable you map

> W <- d$pop      # the comparison variable

> head(d[, c("state", "zone", "pop", "murder")], 6)
               state  zone   pop murder
Alabama       Alabama South  3615   15.1
Alaska         Alaska  West   365   11.3
Arizona       Arizona  West  2212    7.8
Arkansas     Arkansas South  2110   10.1
California California  West 21198   10.3
Colorado     Colorado  West  2541    6.8

> dim(d); nrow(mapdat); length(unique(mapdat$region))
[1] 50 11
[1] 15527
[1] 48
```

Output check: d has 50 rows and 11 columns; the merged map data has 15,527 polygon points for 48 states.

Task 1: The Base Map [4 marks]

Question: Draw a choropleth of murder with ggplot(mapdat, aes(long, lat, group = group, fill = murder)) + geom_polygon(colour = 'white') + coord_map('albers', lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = 'inferno') with a clear title and legend. Report the highest state and its value, the lowest value and the state(s) that have it, and the median of all 50 states.

Program and Output

```
> p1 <- ggplot(mapdat, aes(long, lat, group = group, fill = murder)) +
+   geom_polygon(colour = "white") +
+   coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
+   scale_fill_viridis_c(option = "inferno", name = "Murder rate\n(per 100,000)") +
+   labs(title = "Murder rate by state, 1976 (per 100,000 residents)",
+        subtitle = "48 contiguous states; Alaska and Hawaii are not in map_data",
+        caption = "Source: R datasets::state.x77") +
+   theme_void(base_size = 12) + theme(plot.title = element_text(face = "bold"))

> ggsave("t1_map.png", p1, width = 8, height = 5, dpi = 200, bg = "white")

> d$state[which.max(V)]; max(V)
[1] "Alabama"
[1] 15.1

> d$state[V == min(V)]; min(V)
[1] "North Dakota"
[1] 1.4

> median(V)
[1] 6.85
```

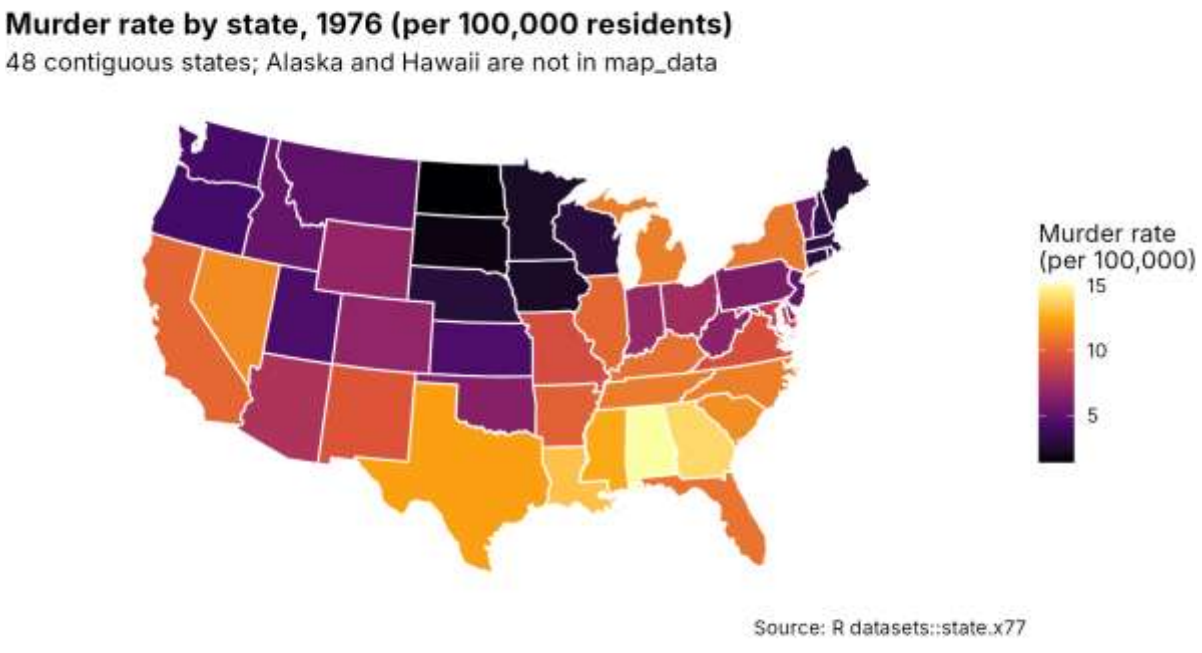


Figure 1: Choropleth of the murder rate by state, 1976 (inferno scale)

Observation

(a) The highest rate is Alabama with 15.1 per 100,000; the lowest is North Dakota with 1.4 (the only state with that value). Alabama's rate is 10.8 times North Dakota's.

(b) The median of all 50 states is 6.85 per 100,000, a little below the mean of 7.38, so a few high states pull the mean up.

(c) On the map the brightest (highest) states form a block in the South-East (Alabama, Georgia, Louisiana, Mississippi), while the darkest (lowest) states are the northern plains (North Dakota, South Dakota, Minnesota, Iowa) and New England.

Result: Highest: Alabama (15.1); lowest: North Dakota (1.4); median of the 50 states: 6.85 per 100,000.

Task 2: The Regional Story [4 marks]

Question: Use tapply(V, d$zone, mean). Give the mean for each of the four regions and name the highest and lowest region and the gap between them. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause.

Program and Output

```
> reg <- tapply(V, d$zone, mean)


> round(reg, 2)
    Northeast         South North Central          West
         4.72         10.58          5.28          7.22


> names(reg)[which.max(reg)]; names(reg)[which.min(reg)]
[1] "South"
[1] "Northeast"


> round(max(reg) - min(reg), 2)
[1] 5.86


> cor(V, W)
[1] 0.3436428


> rd <- data.frame(zone = names(reg), mean_murder = as.numeric(reg))


> p2 <- ggplot(rd, aes(reorder(zone, mean_murder), mean_murder, fill = mean_murder)) +
+   geom_col(width = 0.6) + geom_text(aes(label = round(mean_murder, 2)), vjust = -0.4) +
+   scale_fill_viridis_c(option = "inferno", end = 0.85, guide = "none") + coord_cartesian(ylim = c(0,
12)) +
+   labs(title = "Mean murder rate by region, 1976", x = NULL, y = "Murder rate per 100,000") +
theme_minimal(base_size = 12)


> ggsave("t2_regions.png", p2, width = 6, height = 3.8, dpi = 200, bg = "white")


> p2b <- ggplot(d, aes(pop, murder)) + geom_point(size = 2, colour = "#b5367a") + geom_smooth(method =
"lm", se = FALSE, colour = "grey30", linewidth = 0.6) +
+   labs(title = "Murder rate against population", x = "Population (thousands, 1975)", y = "Murder rate
per 100,000") + theme_minimal(base_size = 12)


> ggsave("t2_scatter.png", p2b, width = 6, height = 3.8, dpi = 200, bg = "white")
```

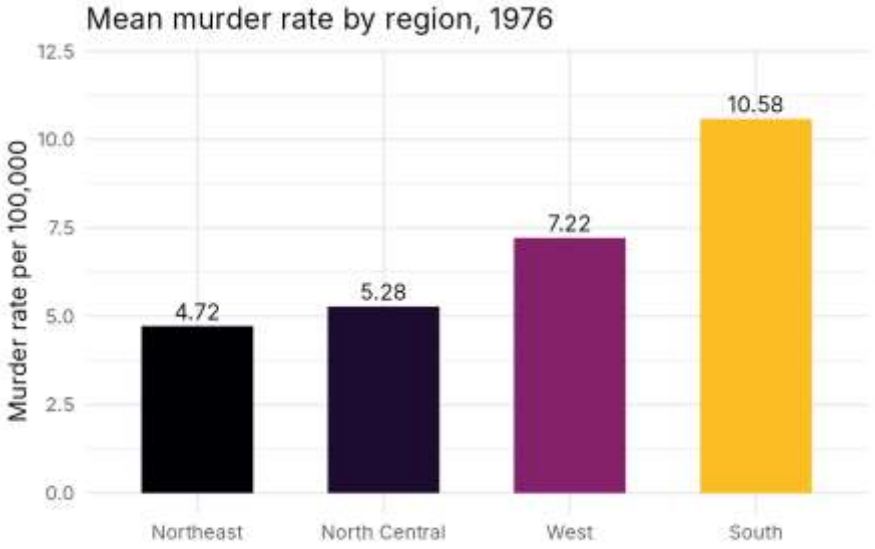


Figure 2: Mean murder rate by region

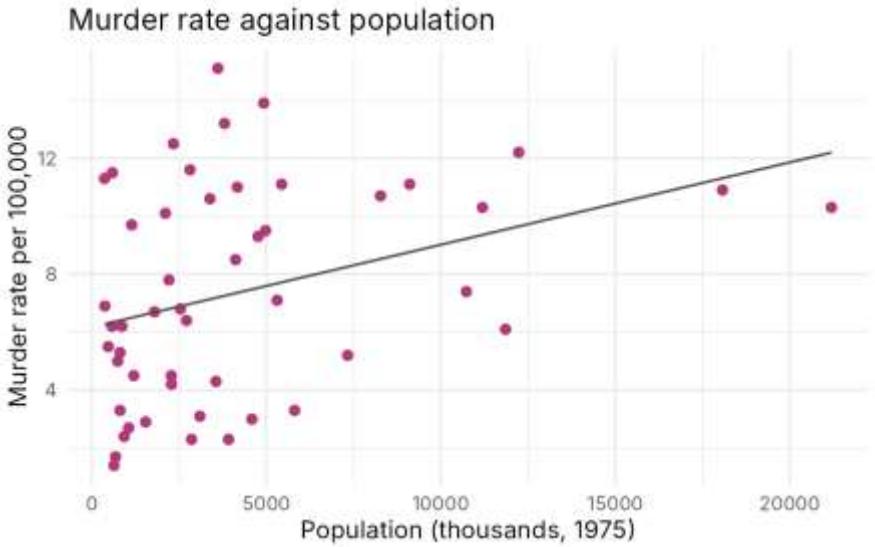


Figure 3: Murder rate against population (50 states)

Observation

(a) Regional means (per 100,000): Northeast 4.72, North Central 5.28, West 7.22 and South 10.58.

(b) The highest region is the South (10.58) and the lowest is the Northeast (4.72); the gap is 5.86 per 100,000, so the South's mean is 2.24 times the Northeast's.

(c) cor(V, W) = +0.344, which is a moderate (0.3 to 0.7) positive correlation: states with more people tend to have somewhat higher murder rates, but population explains only about 12% of the variation (r squared = 0.118) and a correlation is not proof of cause.

Result: The South is the highest region (10.58) and the Northeast the lowest (4.72), a gap of 5.86; cor(murder, population) = 0.344, moderate and positive.

Task 3: Classes Change the Story [4 marks]

Question: Make two versions of the map with 6 classes: equal interval, cut(V, 6), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it. Which method makes the highest look rarer, and which would you choose for your audience?

Program and Output

```
> eq <- cut(V, 6)

> qn <- cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE)

> table(eq)
eq
(1.39,3.68] (3.68,5.97] (5.97,8.25] (8.25,10.5] (10.5,12.8] (12.8,15.1]
         11           8          10           7          11           3

> table(qn)
qn
 [1.4,3.13] (3.13,5.23] (5.23,6.85] (6.85,9.97] (9.97,11.1] (11.1,15.1]
          9           8           8           8           9           8

> levels(eq)[6]; sum(eq == levels(eq)[6])
[1] "(12.8,15.1]"
[1] 3

> levels(qn)[6]; sum(qn == levels(qn)[6])
[1] "(11.1,15.1]"
[1] 8

> sort(V, decreasing = TRUE)[1:12]
 [1] 15.1 13.9 13.2 12.5 12.2 11.6 11.5 11.3 11.1 11.1 11.0 10.9

> d$eq <- eq; d$qn <- qn

> mapdat2 <- merge(map_data("state"), d, by.x = "region", by.y = "id"); mapdat2 <-
mapdat2[order(mapdat2$group, mapdat2$order), ]

> mk <- function(col, ttl) ggplot(mapdat2, aes(long, lat, group = group, fill = .data[[col]])) +
geom_polygon(colour = "white") +
+   coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_d(option = "inferno", name =
"Murder rate", direction = 1) +
+   labs(title = ttl) + theme_void(base_size = 11) + theme(plot.title = element_text(face = "bold"))

> ggsave("t3_equal.png", mk("eq", "Equal-interval classes (cut(V, 6))"), width = 7, height = 4.5, dpi =
200, bg = "white")

> ggsave("t3_quantile.png", mk("qn", "Quantile classes (6 classes, about 8 states each)"), width = 7,
height = 4.5, dpi = 200, bg = "white")
```

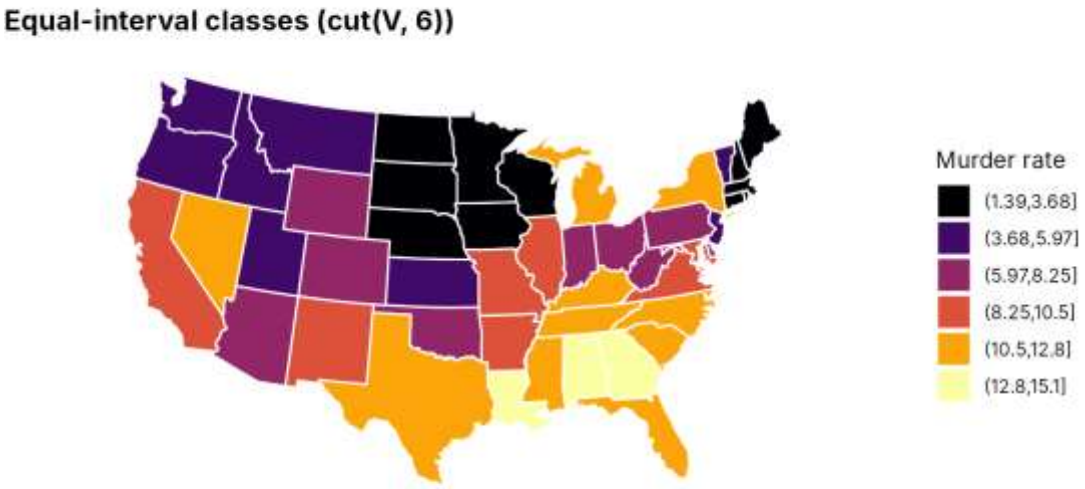


Figure 4: Equal-interval map, 6 classes

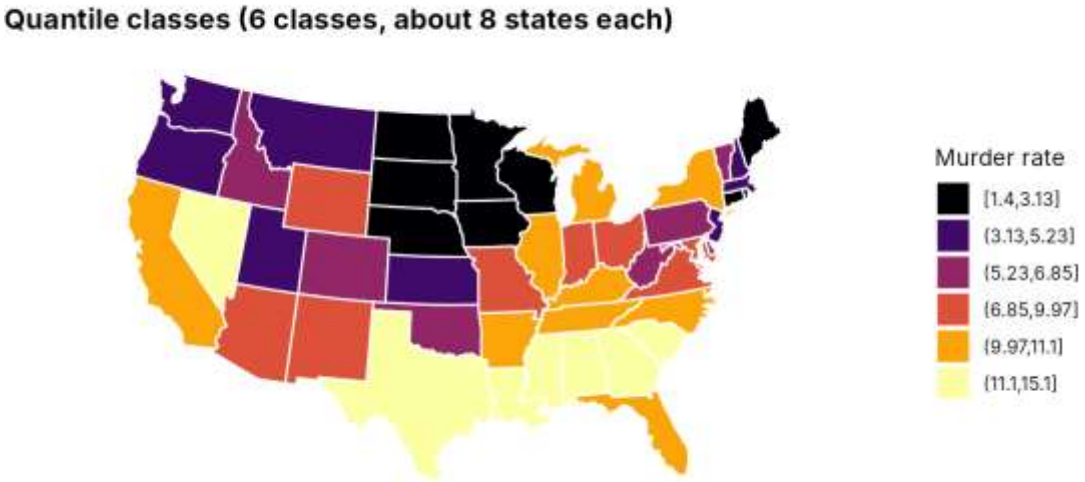


Figure 5: Quantile map, 6 classes

Observation

(a) Equal interval: the six classes are about 2.29 wide and hold 11, 8, 10, 7, 11 and 3 states. The top class is (12.8, 15.1]: its lower limit is 12.8 and it contains 3 states (Alabama 15.1, Georgia 13.9 and Louisiana 13.2).

(b) Quantile: the classes hold 9, 8, 8, 8, 9 and 8 states, almost equal. The top class is (11.1, 15.1]: its lower limit is 11.1 and it contains 8 states.

(c) The equal-interval method makes the highest look rarer: only 3 states (6%) are in the top colour, against 8 states (16%) with the quantile method, although the data are the same.

(d) Choice: for the planning commission I would choose the quantile map. Every colour then covers 8 or 9 states, so the commission can see directly which 8 states are the highest-rate group (rate above 11.1) and plan support for them; the equal-interval map shows the true size of the gaps, but it hides 5 of those 8 states in the next colour (10.5 to 12.8). The class limits (11.1, 9.97 and so on) are less round, so they must be printed in the legend.

Result: Top-class lower limit and number of states: equal interval 12.8 (3 states); quantile 11.1 (8 states). The equal-interval map makes the highest look rarer; the quantile map is chosen for the commission.

Task 4: Counts versus Rates [4 marks]

Question: Create the number of murders, count <- d$pop * 1000 * d$murder / 100000. Name the top 3 states by count and the top 3 by murder rate, and say how many states appear in both lists. In one sentence explain why a map of counts would mislead and what to map instead.

Program and Output

```
> d$count <- d$pop * 1000 * d$murder / 100000


> d[order(-d$count), c("state", "pop", "murder", "count")][1:3, ]
               state   pop murder    count
California California 21198   10.3 2183.394
New York     New York 18076   10.9 1970.284
Texas           Texas 12237   12.2 1492.914


> d[order(-d$murder), c("state", "pop", "murder", "count")][1:3, ]
             state  pop murder   count
Alabama     Alabama 3615   15.1 545.865
Georgia     Georgia 4931   13.9 685.409
Louisiana Louisiana 3806   13.2 502.392


> top_c <- d$state[order(-d$count)][1:3]; top_r <- d$state[order(-d$murder)][1:3]


> intersect(top_c, top_r); length(intersect(top_c, top_r))
character(0)
[1] 0


> sum(d$count[d$state %in% top_c]) / sum(d$count)
[1] 0.3062115


> d$state[which.max(d$pop)]; round(d$count[d$state == "California"] / d$count[d$state == "Alabama"], 2)
[1] "California"
[1] 4


> mapdat3 <- merge(map_data("state"), d, by.x = "region", by.y = "id"); mapdat3 <-
mapdat3[order(mapdat3$group, mapdat3$order), ]


> pc <- ggplot(mapdat3, aes(long, lat, group = group, fill = count)) + geom_polygon(colour = "white") +
coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
+   scale_fill_viridis_c(option = "inferno", name = "Murders") + labs(title = "Counts: number of
murders (mostly shows population)") + theme_void(base_size = 11) + theme(plot.title = element_text(face
= "bold"))


> ggsave("t4_count.png", pc, width = 7, height = 4.5, dpi = 200, bg = "white")
```

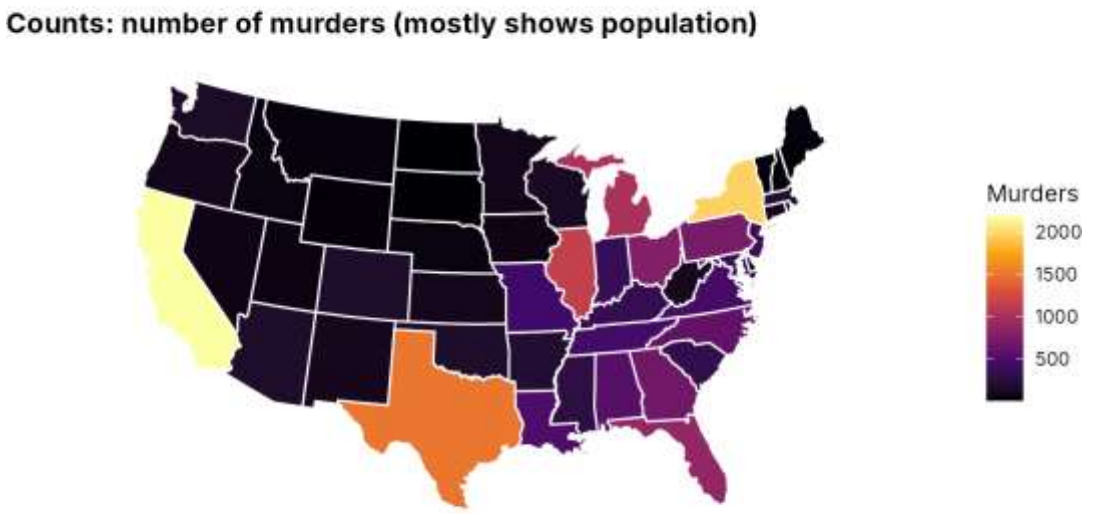


Figure 6: Map of murder counts (compare with the rate map in Figure 1)

Observation

(a) Top 3 by count: California (2,183 murders), New York (1,970) and Texas (1,493).

(b) Top 3 by murder rate: Alabama (15.1), Georgia (13.9) and Louisiana (13.2 per 100,000).

(c) States appearing in both lists: 0 of 3; the two lists are completely different.

(d) Why counts mislead: the count mainly follows population, so the three biggest states hold 30.6% of all the murders but only 24.3% of the people, and California has 4.0 times Alabama's count (2,183 against 546) although its rate (10.3) is lower than Alabama's (15.1). A map of counts shows where many people live, not where the risk is, so the map should show the rate per 100,000 (with counts, if needed, as labels or proportional symbols).

Result: Top 3 by count: California, New York, Texas; top 3 by rate: Alabama, Georgia, Louisiana; 0 states in both. A rate per 100,000 should be mapped, not the count.

Task 5: Tell the Story of the Northeast [4 marks]

Question: Report the mean of murder in the Northeast and its ratio to the national mean of all 50 states; the highest state in the Northeast with its value; and how many Northeast states are above the national median. Then write a headline of at most 12 words, and two evidence sentences plus one recommendation for a state planning commission.

Program and Output

```
> ne <- d[d$zone == "Northeast", ]


> round(mean(ne$murder), 2); round(mean(ne$murder) / mean(V), 2); round(mean(V), 2)
[1] 4.72
[1] 0.64
[1] 7.38


> ne$state[which.max(ne$murder)]; max(ne$murder)
[1] "New York"
[1] 10.9


> sum(ne$murder > median(V)); nrow(ne); median(V)
[1] 1
[1] 9
[1] 6.85


> ne[order(-ne$murder), c("state", "murder", "pop")]
                      state murder   pop
New York           New York   10.9 18076
Pennsylvania   Pennsylvania    6.1 11860
Vermont             Vermont    5.5   472
New Jersey       New Jersey    5.2  7333
Massachusetts Massachusetts    3.3  5814
New Hampshire New Hampshire    3.3   812
Connecticut     Connecticut    3.1  3100
Maine                Maine    2.7  1058
Rhode Island   Rhode Island    2.4   931


> d$focus <- ifelse(d$zone == "Northeast", d$murder, NA)


> mapdat4 <- merge(map_data("state"), d, by.x = "region", by.y = "id"); mapdat4 <-
mapdat4[order(mapdat4$group, mapdat4$order), ]


> cen <- aggregate(cbind(long, lat) ~ region, data = mapdat4[mapdat4$zone == "Northeast", ], FUN =
mean)


> cen <- merge(cen, d[, c("id", "state", "murder")], by.x = "region", by.y = "id")


> p5 <- ggplot(mapdat4, aes(long, lat, group = group)) + geom_polygon(aes(fill = focus), colour =
"white") +
+   geom_text(data = cen, aes(long, lat, label = paste0(state.abb[match(state, state.name)], " ",
murder), group = NULL), size = 2.6, colour = "white", lineheight = 0.9) +
+   coord_map("albers", lat0 = 29.5, lat1 = 45.5, xlim = c(-81, -66.5), ylim = c(39.5, 47.7)) +
+   scale_fill_viridis_c(option = "inferno", limits = c(0, 16), na.value = "grey88", name = "Murder
rate\n(per 100,000)", guide = guide_colourbar(barheight = 6)) +
+   labs(title = "Northeast murder rate is below the national median",
+        subtitle = "Northeast mean 4.72 vs 7.38 for all 50 states; only 1 of 9 states is above the
median 6.85",
+        caption = "Source: R datasets::state.x77 (murder rate per 100,000, 1976). Grey = other
regions.") +
+   theme_void(base_size = 12) + theme(plot.title = element_text(face = "bold"))
```

```
> ggsave("t5_story.png", p5, width = 8, height = 5.2, dpi = 200, bg = "white")

> p5b <- ggplot(ne, aes(reorder(state, murder), murder, fill = murder)) + geom_col(width = 0.65) +
geom_hline(yintercept = median(V), linetype = "dashed") +
+   annotate("text", x = 1, y = median(V) + 0.2, label = paste("national median", median(V)), hjust =
0, size = 3.5) +
+   geom_text(aes(label = murder), hjust = -0.2, size = 3.5) + coord_flip(ylim = c(0, 12)) +
+   scale_fill_viridis_c(option = "inferno", limits = c(0, 16), guide = "none") +
+   labs(title = "Northeast states against the national median", x = NULL, y = "Murder rate per
100,000") + theme_minimal(base_size = 12)

> ggsave("t5_bars.png", p5b, width = 6.5, height = 3.8, dpi = 200, bg = "white")
```

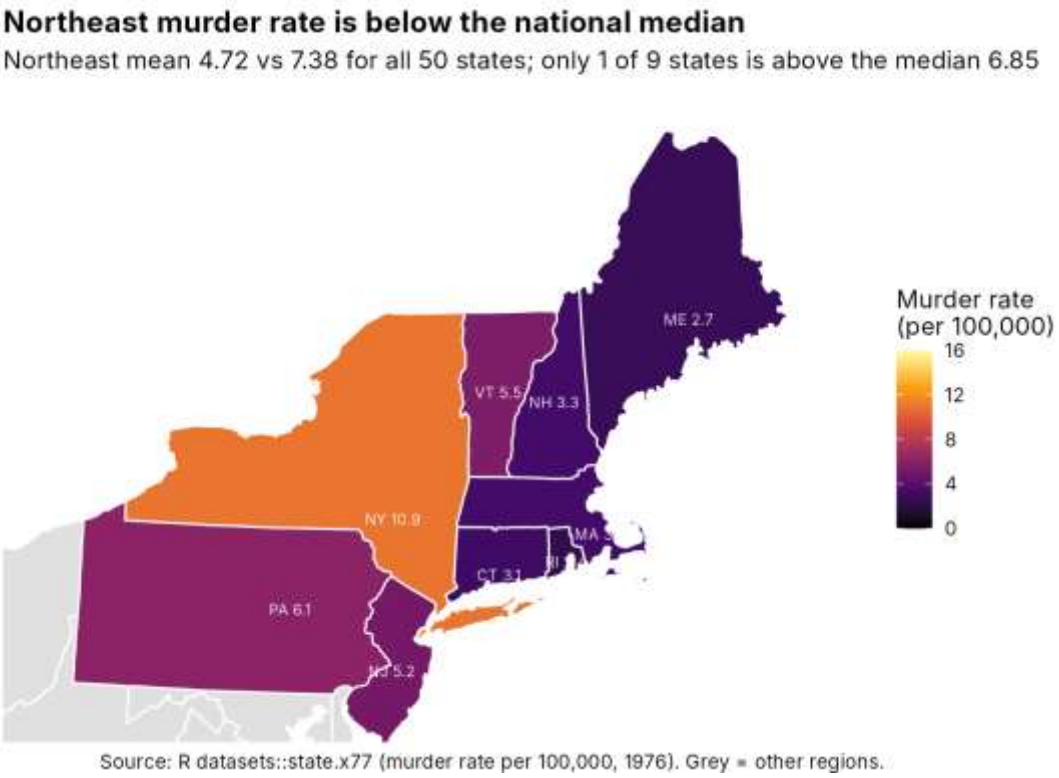


Figure 7: The story map: Northeast states against the rest (other regions in grey)

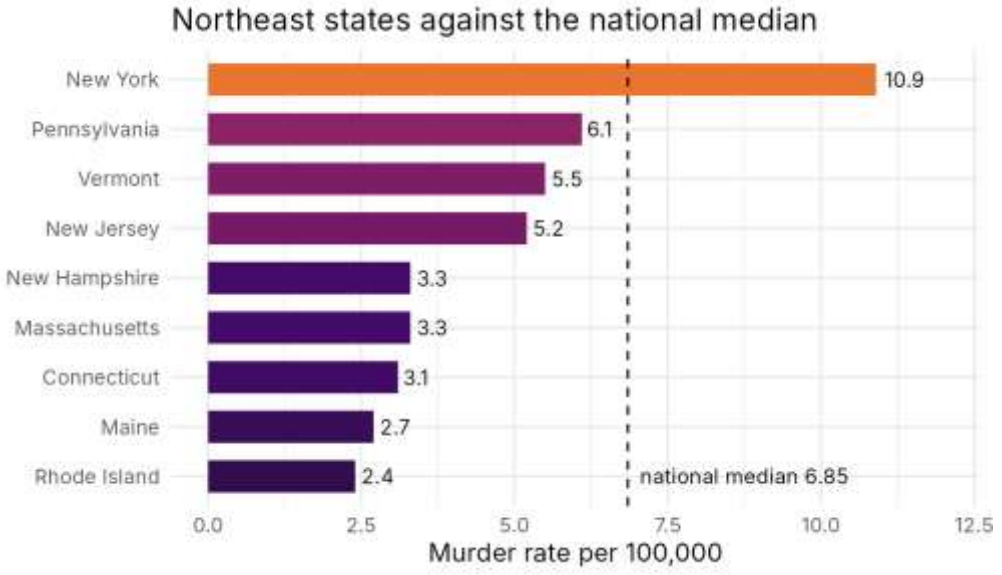


Figure 8: Northeast states against the national median of 6.85

Observation

(a) The Northeast mean is 4.72 per 100,000, which is 0.64 times (36% below) the national mean of 7.38.

(b) The highest state in the Northeast is New York with 10.9; the lowest is Rhode Island with 2.4.

(c) Only 1 of the 9 Northeast states (New York) is above the national median of 6.85; the other 8 are below it.

The Story

Headline (8 words): Northeast murder rate stays below the national median

Evidence 1: The Northeast mean of 4.72 per 100,000 is only 0.64 times the national mean of 7.38, and 8 of its 9 states are below the national median of 6.85.

Evidence 2: Only New York (10.9) is above the median, while Rhode Island (2.4), Maine (2.7) and Connecticut (3.1) are far below it; Rhode Island's rate is just 16% of Alabama's 15.1, the highest in the country.

Recommendation for the planning commission: Treat the Northeast as a low-need region and put the extra prevention funds in the South (mean 10.58 per 100,000), keeping the Northeast effort for New York alone, which also has the second-highest murder count (1,970).

Source line for the map: R datasets::state.x77, murder rate per 100,000 residents, 1976.

Result: Northeast mean 4.72 (0.64 of the national mean 7.38); highest state New York (10.9); 1 of 9 states above the national median (6.85).

Summary of Answers

Task	Main answers
1	Highest Alabama 15.1; lowest North Dakota 1.4; median 6.85
2	Means: Northeast 4.72, North Central 5.28, West 7.22, South 10.58; gap 5.86 (South highest, Northeast lowest); cor = 0.344 (moderate, positive)
3	Equal interval: top class lower limit 12.8, 3 states; quantile: 11.1, 8 states; equal interval makes the highest look rarer; quantile chosen
4	Top 3 by count: California, New York, Texas; top 3 by rate: Alabama, Georgia, Louisiana; 0 in both; map rates, not counts
5	Northeast mean 4.72 = 0.64 x national mean 7.38; highest New York 10.9; 1 of 9 states above the median

Conclusion

A good map story needs one clear message, honest classes and rates instead of raw counts. The executed code shows that the South has the highest murder rate (10.58) and the Northeast the lowest (4.72), that the choice of classes changes how rare the highest group looks (3 states or 8 states), that counts point to California, New York and Texas only because they are populous, and that the Northeast can be told as a story of a low-rate region with one exception, New York.